ALKENE

21. Which one of the following characteristics apply to both ethene and ethyne]
(a) Explode when mixed with chlorine
(b) Decolourise Baeyer's reagent giving brown precipitate
(c) Rapidly absorbed by cold conc. H_2SO_4
(d) Form white precipitate with silver nitrate solution

22. Which of the following has highest knocking property
(a) Aromatic hydrocarbons
(b) Olefins
(c) Branched chain paraffins
(d) Straight chain paraffins

23. Dilute aqueous $KMnO_4$, at room temperature reacts with $R-CH=CH-R$ to give
(a) *R – CHO*
(b) *R – COOH*
(c) *RCHOH – CHOHR*
(d) $CO_2 + H_2O$

24. Aqueous sulphuric acid reacts with 2-methyl-1-butene to give predominantly
(a) Isobutyl hydrogen sulphate
(b) 2-methyl-2-butanol
(c) 2-methyl-1-butanol
(d) Secondary butyl hydrogen sulphate

25. How can ethene be produced from ethanol
(a) By dehydrohalogenation
(b) By dehydrogenation
(c) By dehydration with conc. H_2SO_4 at 170^oC
(d) By reduction with hydrogen iodide

26. Baeyer's reagent is used in the laboratory for
(a) Detection of double bonds
(b) Detection of glucose
(c) Reduction
(d) Oxidation

27. Isopropyl alcohol is obtained by reacting which of the following alkenes with conc. H_2SO_4 and H_2O
(a) Ethylene
(b) Propylene
(c) 2-methyl propene
(d) Isoprene

28. Which of the following compound is produced when $CH_2 = CH - (CH_2)_2COOH$ reacts with *HBr* in presence of peroxides

(a) $CH_3CH(CH_2)_5COOH$

(b) $BrCH_2CH_2(CH_2)_5COOH$

(c) $CH_3CH_2CH_2(CH_2)_5COOH$

(d) $CH_3CH_2BrCH_2CH_2COOH$

29. One mole of each of the following alkenes is catalytically hydrogenated. The quantity of heat evolved will be the lowest in the case of

(a) 1-butene

(b) Trans-2-butene

(c) Cis-2-butene

(d) 1, 3-butadiene

30. Which of the following is not used to distinguish ethene from ethane

(a) Iodine in CCl_4

(b) Bromine in CCl_4

(c) Alkaline $KMnO_4$

(d) Ammonical Cu_2Cl_2

31. A hydrocarbon *X* adds on one mole of hydrogen to give another hydrocarbon and decolourised bromine water. *X* reacts with $KMnO_4$ in presence of acid to give two moles of the same carboxylic acid. The structure of *X* is

(a) $CH_2 = CH - CH_2CH_2CH_3$

(b) $CH_3CH_2CH_2 - CH = CHCH_3$

(c) $CH_3CH_2CH = CHCH_2CH_3$

(d) $CH_3CH = CHCH_2CH_2CH_3$

32. When 2-bromobutane reacts with alcoholic *KOH*, the reaction is called

(a) Halogenation

(b) Hydrogenation

(c) Chlorination

(d) Dehydro-halogenation

33. 1, 3-butadiene reacts with ethylene to form

(a) Benzene

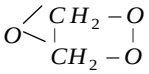
(b) Cyclohexane

(c) Cyclohexene

(d) 2, 3 dimethyl butane

34. Ethylene reacts with ozone gas to form the compound

(a) $HCHO$ (b) C_2H_5OH

(c)  (d) CH_3CHO

35. Oils are converted into fats by

(a) Hydration

(b) Decarboxylation

(c) Hydrogentation

(d) Dehydrogenation

(e) Hydrogenolysis

36. Which process converts olefins into parafins]

(a) Halogenation

(b) Dehydration

(c) Hydrogenation

(d) Hydrolysis

37. Of the following the formula which represents a saturated cyclic compound is

(a) C_3H_6 (b) C_3H_8

(c) C_8H_{10} (d) C_8H_{12}

38. In a reaction, if half of the double bond is broken and two new bonds are formed, this is a case of

(a) Elimination

(b) Addition

(c) Displacement

(d) Rearrangement

39. Which of the following are formed on addition reaction of DCI with 3-methyl-1-butene

(a) $CH_2DCHClCH(CH_3)_2$

(b) $CH_2DCH_2CCl(CH_3)_2$

(c) $CH_3CDClCH(CH_3)_2$

(d) $ClCH_2CHDCH(CH_3)_2$

40. Major product of the following reaction is

$$CH_3-\overset{\overset{Br}{|}}{\underset{\underset{H}{|}}{C}}-CH_2-CH_3 + alco.KOH \rightarrow$$

(a) Butene-1 (b) Butene-2

(c) Butane (d) Butyne-1

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